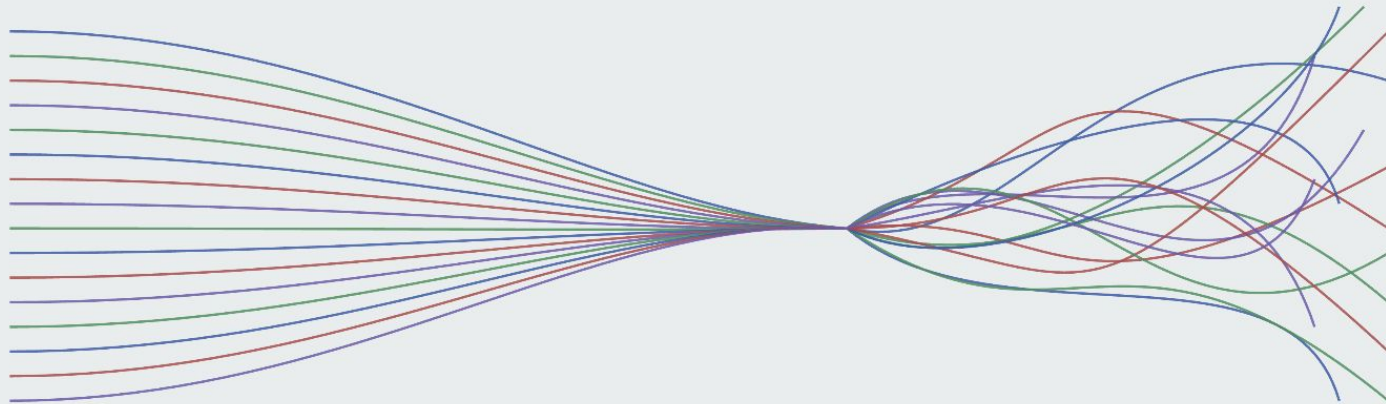


Universal Aesthetics: A Multimodal Perspective

Do AI Models Share a Common Sense of Beauty?

Beauty

Ugliness



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Motivation & Core Hypothesis

“All happy families are alike;
each unhappy family is unhappy
in its own way.”

— *Anna Karenina*, Lev Tolstoy

“All beautiful things are alike; each
ugly thing is ugly in its own way.”

— Anna Karenina Aesthetic
Principle



For AI models

Goal: Test whether “aesthetic” texts/images show **higher convergence** across AI models.



Research Framework



Two stages:

1. **Language Fusion:** Compare poetry vs plain/descriptive text
2. **Multimodal Fusion:** Compare image-text vs image-poem pairs

Key Questions

1. Do LLMs **converge** more on poetry than plain text?
2. What **drives** this convergence? (Semantics? Rarity?)
3. Does image **aesthetics** affect cross-modal alignment?





Data Preparation

1. Plain Text: An open-source Hugging Face Dataset (from Huh et al.'s paper)
2. Poetry: An open-source Kaggle Poems Dataset → filtered & cleaned
3. **Aesthetic Scoring:**
 - a. 5-point scale (1 = ugly, 5 = beautiful)
 - b. Scored by GPT-4o using CoT + weighted criteria (imagery 30%, emotion 25%, etc.)
4. Image-Poem Data: ~1,000 images → poems generated via VGG + seq2seq model (from Liu et al.'s paper); 5-point scale aesthetics rating for images (from Ma et al.'s paper)



Measuring Convergence

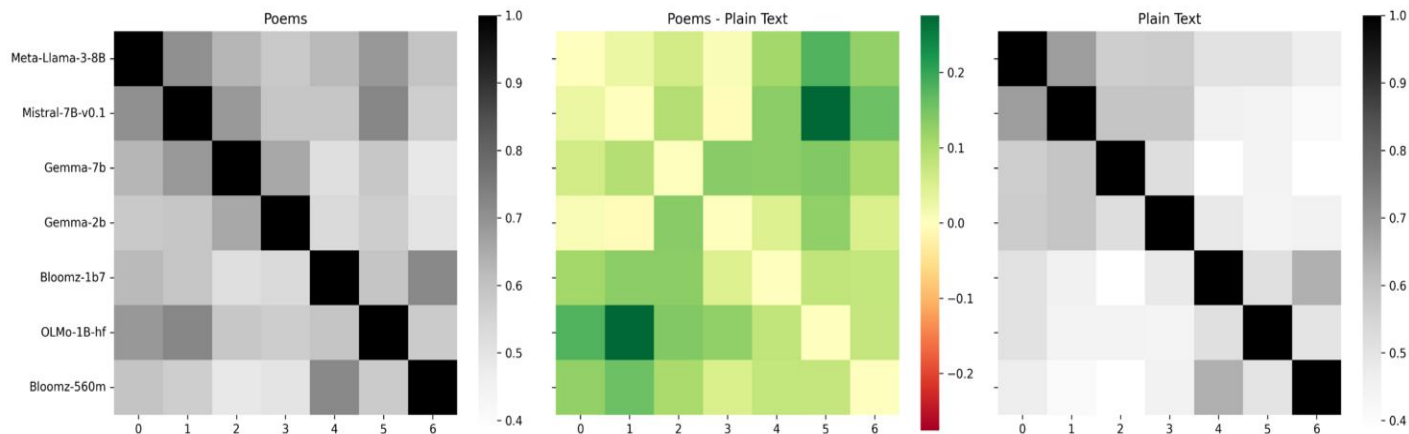
1. Model selection: 7 LLMs (Llama-3-8B, Mistral-7B, Gemma-7B/2B, BLOOMZ-1.7B/560M, OLMo-1B); cover diverse architectures, scales, and training paradigms

2. Metric: Mutual k-Nearest Neighbors; for any two models, compute overlap of nearest neighbors in representation space

Higher overlap → Higher convergence

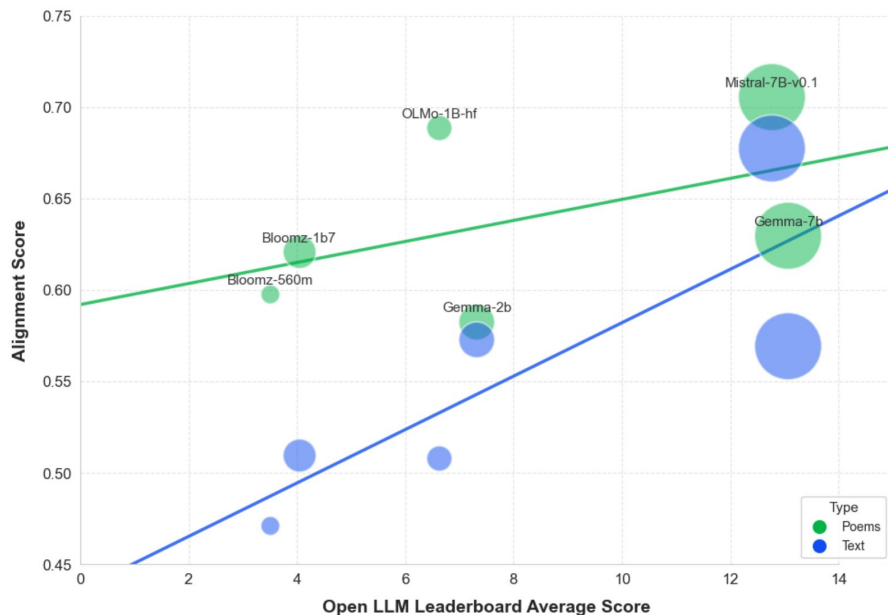
3. Visualization: Heatmaps, t-SNE, model capability

The First Clue: Models Converge More on Poems



Across all model pairs, representation spaces are **significantly more aligned** when processing poems compared to plain text. This confirms our initial hypothesis in the language domain.

Capability and Alignment



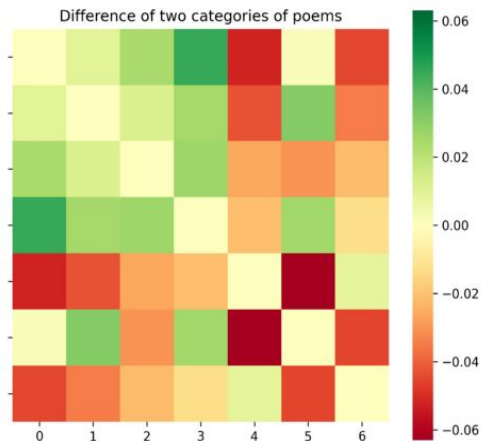
Convergence: It's stronger for poems across the board.

Trend: More capable models produce more similar representations → They are all learning a **shared, underlying structure**.

Pattern: The trend for **poems** is flatter → smaller models already achieve relatively good alignment for poems.

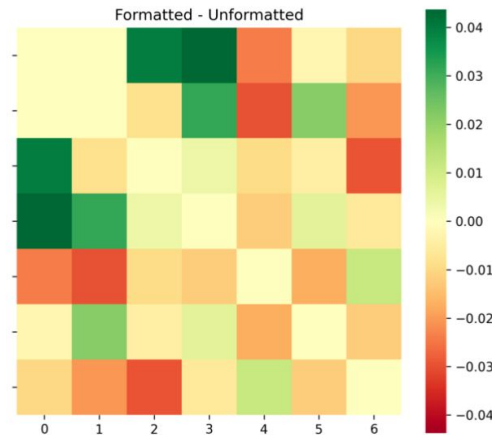
Fine-Grained Aesthetics: Score and Format

High Aesthetic Score - Low Aesthetic Score



Model convergence on high score poems isn't consistently higher; large models converge slightly better on high-score poems.

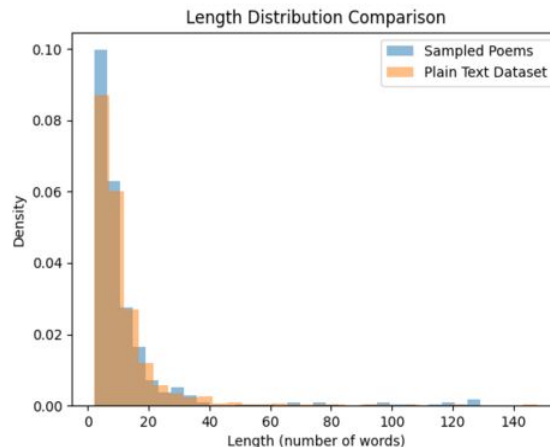
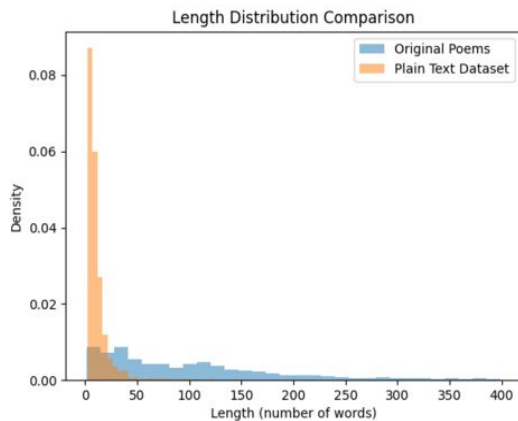
Formatted - Unformatted



Large models prefer structured poems. Small models prefer free verse.

Simple aesthetic labels are insufficient. A deeper mechanism is at play, and it appears to affect large and small models differently.

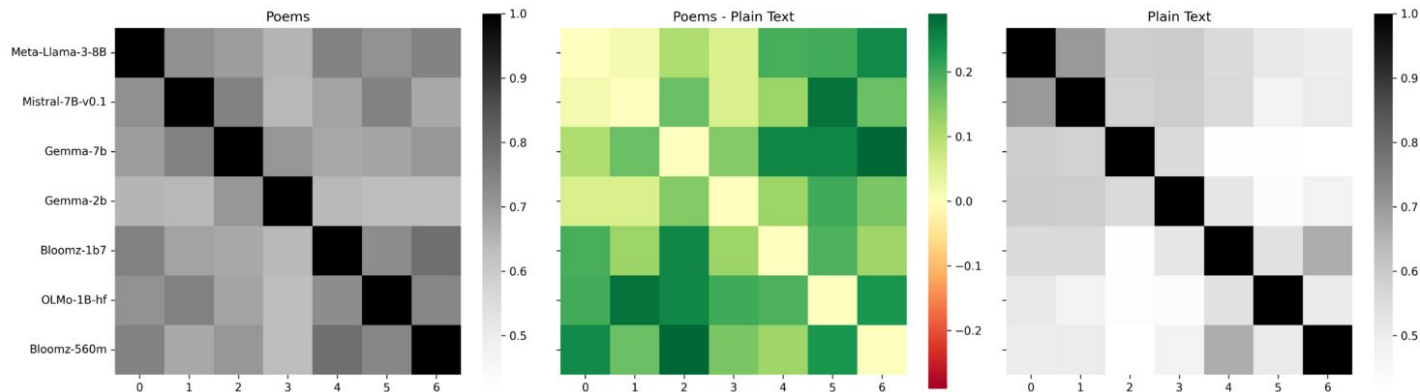
Controlling for Length Bias



Poems are longer/shorter → **potential confounder**

Solution: Resampling poems to match plain-text length distribution; training a logical classifier in a shared PCA space and using predicted probabilities to resample plain texts that match poem representations

Controlling for Length Bias: Result

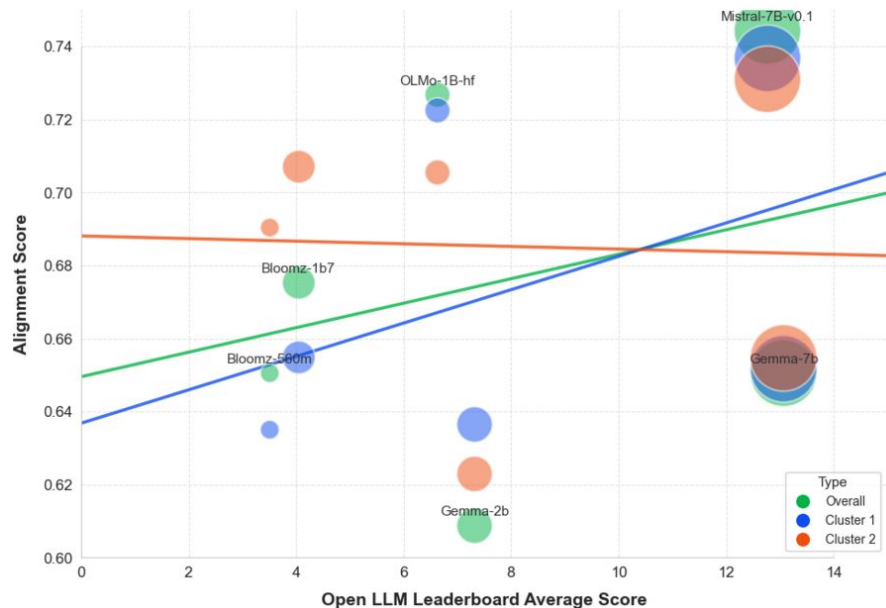


Smaller models show more **obvious convergence advantage** on the poem dataset according to the distribution after resampling.

Hypothesis: Aesthetic features can be **learned early** in model training.

Small models' convergence is more sensitive to aesthetic quality and text simplification. But they struggle to capture fine-grained aesthetic differences.

Behind Aesthetics: Semantic Features

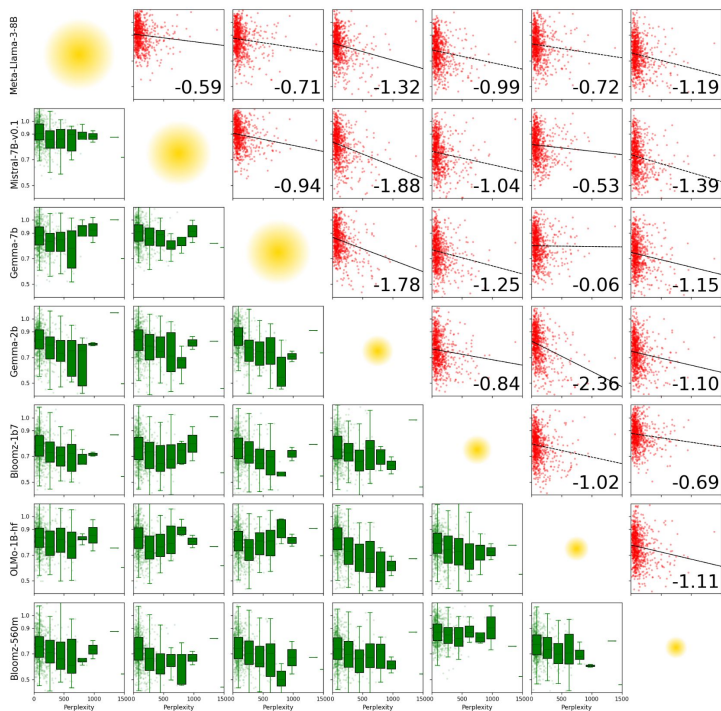


Convergence patterns differ between large and small models: overall, the alignment is lower between semantically similar data (more difficult to distinguish)

Contrary to convergence difference between poems and plain texts

Great difference in scores between two clusters for small models: **greater difficulty distinguishing the fine-grained aesthetic differences within the poetry dataset.**

Behind Aesthetics: Rarity



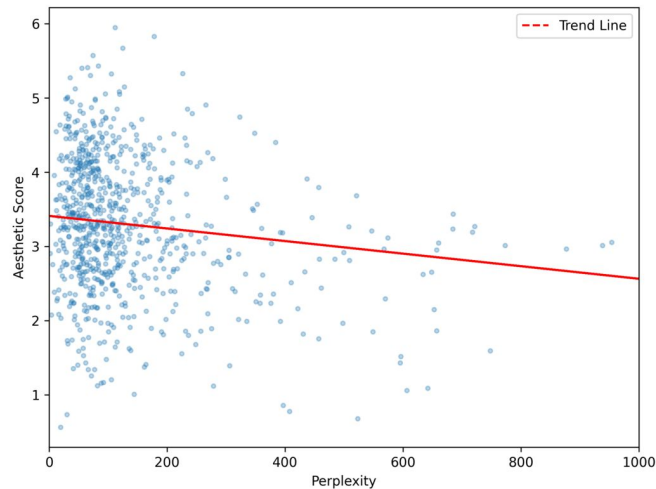
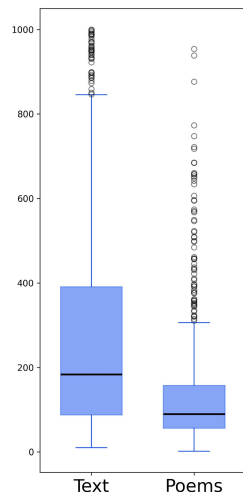
Perplexity

A measure of how 'surprised' a model is by a text. High perplexity means the text is rare or unfamiliar to the model; low perplexity means it is common.

There is a **consistent and strong negative correlation** between model convergence and perplexity.

Models converge on **common, familiar patterns**.

Behind Aesthetics: Rarity



What we perceive as 'aesthetic' is not complex or rare to a language model.

Instead, it corresponds to **fundamental, well learned patterns** in the training data.

Beauty is the mastery of the basics for AI.

Extends to Vision: Image-Text Pairs

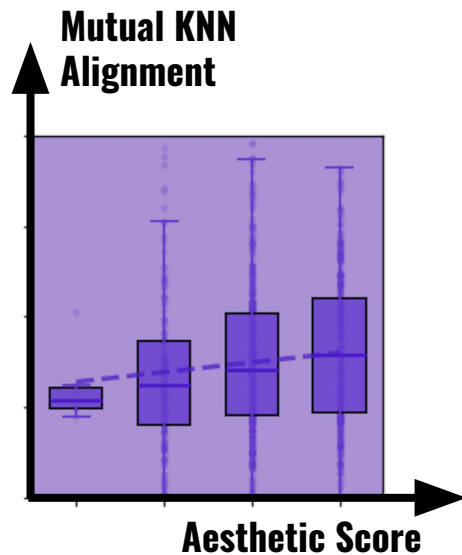
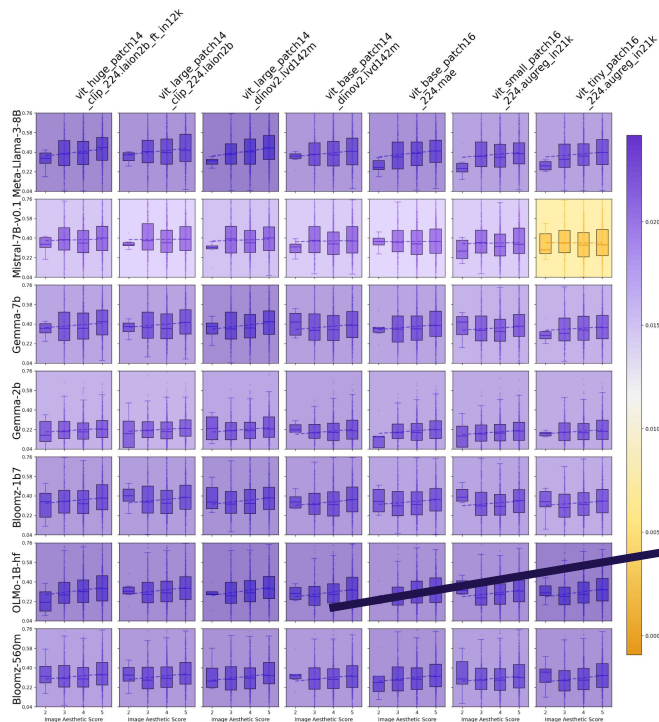
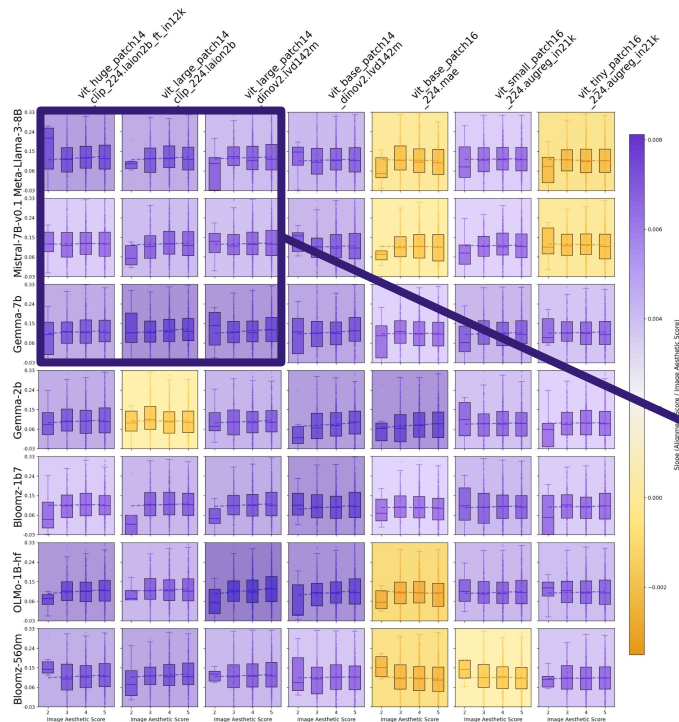


Image-text pairs:
positive correlation
between
convergence and
image aesthetics

Models also converge on beautiful images!

Image-Poem Pairs



Models still perform better on beautiful images, but effects are **weaker**

Might be influenced by the **huge semantic dissimilarity** of the poems and the images

Safe Area - Large Model Pairs

Indicating small models struggle more to capture **fine aesthetic differences**, especially in image-poem pairings



What We Discovered About AI and Beauty

- **AI models share a sense of beauty**

Diverse models from different architectures and modalities consistently agree more on the representations of aesthetically pleasing data (poems, images).

- **"Aesthetic" features are fundamental, not sophisticated**

Convergence is driven by low rarity (low perplexity). What we call beautiful corresponds to common, core patterns that models master early in training.

- **Model capability determines fine-grained perception**

While all models learn these fundamental patterns, larger, more capable models are better at distinguishing nuanced aesthetic differences, a trend observed in both language and vision.



Limitations and Future Directions

Contributions

Dataset

- ~800 Poems with aesthetic scores
- ~1000 Image-Description Pairs with poems and aesthetic scores for images

Source Code:

- Full repository containing both experiments and plotting scripts

Limitations

Constraint Scope

- Limited to models $\leq 8B$
- Dataset size and variety are limited
- Cross-modal convergence experiments are exploratory

Potential Biases:

- Using GPT-4o for aesthetic scoring
- Mutual kNN cannot fully compare across datasets


Thank you!

